

Ahmed Ikram | AI Engineer

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Final-year BSc Computer Science (Data Science & AI), Dundee (predicted First, June 2027). Built agentic-RAG, LLMops and ML systems in Agile delivery, including an NCR Atleos industry project with weekly stakeholder demos - translating progress for an engineer and a non-technical PM, shaping the build on feedback. Self-taught Terraform to production standard. Mentored junior pupils one-to-one. Seeking a graduate AI Engineering role from June 2027.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a First-Class Honours degree. Expected Graduation: June 2027

High School of Dundee

Sep 2018 - June 2023

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

SKILLS

Languages & Data: Python · SQL · TypeScript · JavaScript · Bash · Rust / PyO3 · PostgreSQL · Redis · Kafka · Pydantic

LLMs: PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · AWQ · Unsloth · TRL · Accelerate · BitsandBytes · vLLM · Jinja2, Prompt & Context Engineering

Agents & RAG: LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · Guardrails

Evaluation & Observability: RAGAS · Execution-Based Scoring · LLM-as-Judge · LangSmith · Langfuse · Evidently AI

MLOps & Serving: MLflow · Weights & Biases · Apache Airflow · FastAPI · GraphQL · Modal · AWS SageMaker

Classic ML: XGBoost · Isolation Forest · scikit-learn · Optuna · Tesseract · OpenCV

Cloud: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · Google Cloud (GCS) · Terraform · Docker · GitHub Actions / CI/CD · Linux

PROJECTS

ATM Log Aggregation, Anomaly Detection & Diagnostics - RAG over Logs and ML Anomaly Detection (NCR Atleos)

LangGraph · MCP · ChromaDB · FastAPI · XGBoost · MLflow · Kafka · PostgreSQL · Redis · Docker · Terraform · AWS

- Built a Hybrid RAG as a LangGraph agent over a 12-tool MCP server with semantic chunking & local embeddings, cross-encoder reranking, grounding-gated re-retrieval and uncertainty-weighted 4-signal confidence for refusal and citation (hallucination control).
- Measured 3 retrieval strategies (baseline, hybrid, agentic) with RAGAS on a 50-query golden set: adaptive routing with re-retrieval nearly doubles context (0.708 vs 0.350) at the best context precision (0.874); a CI gate blocks any merge regressing a RAGAS metric by >0.05.
- Architected three-layer anomaly detection (XGBoost + Isolation Forest, Z-score, 7-heuristics) on a live Kafka Raft stream - 0.70 F1 at 97.3% precision via a 200-point Youden sweep - versioned in MLflow with champion aliases and cross-checked against a live SageMaker endpoint.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager, 4 least-privilege IAM roles, state locked in DynamoDB) behind OIDC CI/CD, validated by 1,438 tests across pytest, vitest, playwright and Terraform assertions.

StockLens FinTech Spending & Investment App - AI Agent, Multimodal Extraction, MLOps

LangGraph · MCP · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · GraphQL · Redis · Terraform

- Built a 2-node ReAct agent (StateGraph, not `create\_react\_agent` wrapper) exposing 16 tools to AWS Bedrock via a Stateless MCP server (Streamable HTTP, OAuth 2.1 PKCE). Correctness judged in CI by an LLM-as-judge with LangSmith tracing.
- Shipped a confidence-gated receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM fallback) that only spends LLM budget when OCR confidence drops, matched to merchant with rapidfuzz.
- Trained a global LSTM (entity embeddings, Optuna 30-trial HPO, focal loss) reaching 51.6% five-day directional accuracy vs a 33% 3-class baseline (0.75 simulated Sharpe). 13 features computed in a Rust/PyO3 engine; retrained weekly via Airflow.
- Promotion requires beating the live champion by >2pp, gated by a binomial significance test ($p < 0.05$), with Evidently drift monitoring (PSI/KS/JSD); shipped on a Terraform-provisioned AWS estate (ECS Fargate, EFS champion serving) through 9 parallel CI jobs and an 8-job, 5-stage OIDC CD with 2,427 tests at a 90% coverage gate.

SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving

Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud

- Architected a model-agnostic LLMops platform: one CLI command onboards any open-weight Hugging Face model into the full pipeline - schema-validated data prep → QLoRA fine-tuning → execution-based evaluation → statistically-gated promotion → config-driven OpenAI-compatible serving; versioned GCS/W&B artifacts shared across every stage, with per-generation traces in Langfuse.
- Engineered a six-stage leak-free data pipeline: 20,477 raw SWE-bench issues → 17,456-examples across 46 repos; each repo held in exactly one split.
- Evaluation against the ground truth: every task materialises its repo in the official Docker image and executes the genuine pass/fail suite for ground-truth scoring.
- Fine-tuned 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth, FlashAttention on a Modal A100); champion hit 17.20% fail-to-pass on a 100-instance golden sample (7× base, 95% CI 11.1-25.8%, McNemar $p \approx 6e-05$ and 90.10% unrelated pass-pass rate, +61.6 pt vs base); promoted only past a 4-condition statistical gate.
- Served on Modal scale-to-zero with vLLM & per-request LoRA swapping and AWQ-quantisation or full-precision per the model registry; a 20-instance F2P smoke gate blocks regressing PRs. GCP via Terraform and Workload Identity Federation; 1,539 tests.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
<i>Retail Assistant</i>	<i>Sep 2023 - Present</i>
Zulus Peri Peri Grill , Dundee, Scotland	
<i>Front of House Worker</i>	<i>June 2021 - Sep 2023</i>